DAVINCI Architect

Multi-Agent Software Architect

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SYSTEM: 7ep

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow librarian registration using username and password. - The system must not allow registering a librarian with an already existing username. - The system must require filling in username and password during registration. - The system must accept only passwords considered strong enough for safe use. - The system must reject very short passwords. - The system must allow authenticated librarians to provide username and password to enter the system. - The system must grant access only when username and password exactly match the existing registration. - The system must deny access when username or password are incorrect or not provided. - The system must handle librarian credentials securely, without depending on plain-text passwords for later validation. - The system must respond clearly to denied access attempts, without exposing sensitive internal processing details. - The system must maintain authentication as a separate capability, responsible for registration, login, and session validation. - The system must allow registering books in the collection by title. - The system must not allow duplicate registration of the same book. - The system must inform when book registration fails due to missing title. - The system must allow deleting already registered books. - The system must not allow deleting books that were never registered. - The system must also remove the loan link when a removed book is currently on loan. - The system must allow listing all registered books. - The system must allow consulting a book by identifier. - The system must allow consulting a book by title. - The system must allow listing only books available for loan. - The system must inform when there are no registered books or when a search returns no results. - The system must maintain the book catalog as a separate capability. - The system must allow registering readers eligible for loans. - The system must not allow duplicate registration of the same reader. - The system must inform when reader registration fails due to missing name. - The system must allow deleting already registered readers. - The system must not allow deleting readers that were never registered. - The system must also remove loan links when a removed reader has borrowed books. - The system must allow listing all registered readers. - The system must allow consulting a reader by identifier. - The system must allow consulting a reader by name. - The system must inform when there are no registered readers or when a search returns no results. - The system must not accept simultaneous search by identifier and by name/title in the same query. - The system must maintain reader management as a separate capability. - The system must allow registering a book loan for a registered reader. - The system must record the loan date. - The system must not allow loans for people not registered as readers. - The system must not allow loans for unregistered books. - The system must not

allow loans for books already borrowed by another person. - The system must allow the same reader to borrow multiple books at the same time. - The system must ensure the same book is loaned to only one reader at a time. - The system must clearly inform when a loan cannot be completed. - The system must consult reader and book data during loan registration, ensuring information is current. - The system must preserve consistency among books, readers, and loans, avoiding invalid or contradictory records. - The system must maintain loan management as a separate capability. - The loan capability depends on the reader management capability to validate the reader. - The loan capability depends on the book catalog capability to validate the book. - The system must allow clearing data and base structure to reset the environment. - The system must allow recreating the base structure when necessary. - The system must allow quickly restoring the environment to a clean state ready for new tests. - To perform these reset operations, the system must coordinate the involved functional areas, ensuring data is restored consistently. - The system must maintain environment management as a separate capability. - The system must allow summing two integers informed by the user. - The system must allow calculating Fibonacci sequence values using more than one calculation approach. - The system must allow calculating Ackermann function results using more than one calculation approach. - The system must inform an error when mathematical services receive values outside the expected format. - The system must offer a service catalog for consulting and testing available endpoints. - The system must maintain demo utility as a separate capability. - The system must record main operations to support traceability and problem analysis. - To achieve this, relevant operations from registration, authentication, consultation, and loan flows must be reported to a central event recording mechanism. - The system must maintain this record decoupled from business areas, allowing failure and behavior tracking without coupling audit persistence to functional domains. - The system must maintain audit as a separate capability. - The authentication, book catalog, reader management, and loan capabilities depend on the audit capability to record their main operations. - The system must be usable through simple web forms, with fast and readable feedback for each action. - The system must offer clear success, denial, and error messages. - The system must reduce manual typing in repetitive flows, such as book loans, through lists or suggestions. - The system must facilitate loan completion by showing available books and registered readers. - The system must adapt the loan selection method according to the number of available options. - The system must block the loan selection field when no data is available. - The system must display the result of each operation immediately after form submission. - The system must maintain consistent behavior across the main interface and data query/sending endpoints. - The system must work in common browsers and consider small screen usage. - Central registration, authentication, consultation, and loan rules must be repeatably verifiable by automated tests. - The system must be organized by business domains, supporting maintenance and feature expansion. - The code implements book and reader deletion rules, but the current main screen does not directly expose these commands. - The system has librarian authentication, but the current code does not show session blocking in other library operations. - The library and authentication part is

the clearest business core; mathematical modules appear as auxiliary demonstration features.

Reference Services

Authentication Service Book Catalog Service Reader Management Service Loan Management Service Environment Management Service Demo Utility Service Audit Service

Reference Interactions

Authentication Service <-> Audit Service Book Catalog Service <-> Loan Management Service Book Catalog Service <-> Audit Service Reader Management Service <-> Loan Management Service Reader Management Service <-> Audit Service Loan Management Service <-> Audit Service Environment Management Service <-> Authentication Service Environment Management Service <-> Book Catalog Service Environment Management Service <-> Reader Management Service Environment Management Service <-> Loan Management Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Authentication Service,Register librarians with username and password;Validate password strength;Authenticate li
Book Catalog Service,Register books in the collection by title;Prevent duplicate book registration;Delete regist
Reader Management Service,Register readers eligible for loans;Prevent duplicate reader registration;Delete regis
Loan Management Service,Register book loans for registered readers;Record loan dates;Ensure a book is loaned to
Audit Service,Record central operations from registration authentication consultation and loan flows;Support tra
Environment Management Service,Clear data and base structure to reset the environment;Recreate base structure;Co
Demo Utility Service,Sum two integers;Calculate Fibonacci sequence values using multiple approaches;Calculate Ac

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Authentication Service,Register librarians with username and strong password;Authenticate librarians and validat
Book Catalog Service,Register books in the catalog by title;Delete registered books;Consult and list registered
Reader Management Service,Register readers eligible for loans;Delete registered readers;Consult and list registe
Loan Management Service,Register book loans for registered readers with loan date;Enforce single-borrower and bo
Environment Management Service,Reset system data and base structure;Recreate base structure;Coordinate involved
Demo Utility Service,Sum two user-provided integers;Calculate Fibonacci sequence values using multiple approache
Audit Service,Record central operations from registration,authentication

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

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Consolidated Architecture CSV

Microservice,Responsibilities,Communicates With

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Environment Management Service,Clear data and base structure to reset the environment;Recreate base structure;Co
Demo Utility Service,Sum two integers;Calculate Fibonacci sequence values using multiple approaches;Calculate Ac

SYSTEM: acmeair

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow customers to access their account using login and password. - The system must create a usage session after successful authentication. - The system must allow customers to end their session through logout. - The system must require an active session to access restricted areas, such as customer profile and bookings. - The system must restrict private functionalities to authenticated customers. - The customer session must have limited validity and expire automatically after the configured period. - Logout must invalidate the current session immediately. - Customer personal data must be protected against improper access by other customers. - The system must maintain authentication as a separate capability, responsible exclusively for login, session creation, active session validation, and logout. - The system must allow flight search by informing origin airport, destination airport, and travel date. - The system must allow one-way or round-trip searches. - The system must present the flight options found for each trip segment. - For each flight found, the system must show departure time, arrival time, and reference fare values. - The system must work with a set of previously registered airports. - Flight searches must prioritize fast response, as they represent a recurring customer journey flow. - The system must maintain flight search as an independent capability, responsible for airports, routes, schedules, and reference fares. - The system must allow an authenticated customer to create a booking for a one-way flight. - The system must allow an authenticated customer to create a booking for outbound and return flights in the same operation. - The system must generate an identifier for each created booking. - The system must record the booking date. - The system must inform the customer of the booking identifier after operation completion. - The system must allow customers to consult existing bookings. - The system must present the list of completed bookings. - The system must allow canceling a selected booking. - After cancellation, the booking must no longer appear as active in the customer listing. - The booking capability depends on customer authentication. - The booking capability depends on flight search to obtain available flights. - The system must allow customers to consult their own profile. - The profile must show customer contact data, including phone, phone type, and address. - The system must allow updating contact and address data. - The system must require the current password to confirm profile changes. - The system must prevent customers from viewing or changing another customer profile. - The system must maintain the customer profile as a separate capability from authentication, responsible only for registration and contact data. - The customer profile capability depends on authentication to validate customer access. - The system must show the customer loyalty program status. - The system must show the number of miles accumulated

in the year. - The system must show the total accumulated miles. - Loyalty information must be presented as read-only data, not as editable fields. - The system must maintain loyalty as a separate capability from the customer profile, responsible exclusively for status and accumulated miles. - The loyalty capability depends on authentication to validate customer access. - The system must allow initial loading or reloading of the application database. - The loading process must allow defining the number of sample customers to be generated. - The database load must recreate the main operational data, including flights, customers, sessions, and test bookings. - The system must provide summarized operational information, such as numbers of customers, sessions, flights, segments, bookings, and registered airports. - The environment must offer simple ways to prepare and verify the database for tests, demonstrations, or operational reset. - The system must provide basic operational indicators without requiring direct database access. - The system must maintain environment administration as a separate capability, responsible for preparing, reloading, and supervising operational data. - The environment administration capability depends on authentication, flight search, booking, and customer profile to prepare and reset their data. - The application must support high call and query volume, especially in authentication, flight search, and booking flows. - The solution must be suitable for growth without manual support for main self-service operations. - The application must run in different environments, including local infrastructure, public cloud, and hybrid scenarios. - The solution must keep customer, flight, session, and booking data persisted between requests. - Each session and booking must have a unique identifier. - Booking and cancellation operations must maintain enough consistency to correctly reflect the current booking state. - The application must record minimal operational tracking data, such as booking date and operational record counts. - Navigation must support self-service, allowing customers to search flights, manage bookings, and update their profile without human intermediation. - The system must show schedules, dates, and values clearly. - Main journeys must be accessible through a simple web interface, with clear separation among flight search, bookings, profile, and support. - No public customer registration flow was identified. Customer accounts are generated by administrative data loading. - No payment flow, seat selection, or formal fare class selection was identified at booking time; the system shows prices, but the observed booking is simplified. - A Support interface with real-time chat appearance exists, but the corresponding backend was not found. Therefore, this item was not treated as a consolidated functional requirement.

Reference Services

Authentication Service Flight Search Service Booking Service Customer Profile Service Loyalty Service Environment Administration Service

Reference Interactions

Booking Service <-> Authentication Service Booking Service <-> Flight Search Service Customer Profile Service <-> Authentication Service Loyalty Service <-> Authentication Service Environment Administration Service <-> Authentication Service Environment Administration Service <-> Flight Search Service Environment Administration Service <-> Booking Service Environment Administration Service <-> Customer Profile Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Authentication Service,Authenticate customers using login and password;Create usage sessions upon successful aut

Flight Search Service,Search one-way and round-trip flights by origin airport destination airport and travel dat

Booking Service,Create bookings for one-way and round-trip flights;Generate unique booking identifiers and recor

Customer Profile Service,Retrieve customer contact and address data;Update customer contact and address informat

Loyalty Service,Retrieve customer loyalty program status;Display accumulated miles in the year;Display total acc

Environment Administration Service,Load and reload database with configurable sample customer records;Recreate o

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Authentication Service,Authenticate customers via login and password;Create and maintain active customer session

Flight Search Service,Maintain registered airports and routes;Search one-way and round-trip flights by origin de

Booking Service,Create flight bookings for one-way and round-trip itineraries;Generate unique booking identifier

Customer Profile Service,Retrieve customer contact details and address information;Update customer contact and a

Loyalty Service,Track and display customer loyalty program tier status;Provide annual accumulated miles;Provide

Environment Administration Service,Execute initial loading and reloading of application operational data;Generat

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

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Consolidated Architecture CSV

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Customer Profile Service,Retrieve customer contact and address data;Update customer contact and address informat

Loyalty Service,Retrieve customer loyalty program status;Display accumulated miles in the year;Display total acc

Environment Administration Service,Load and reload database with configurable sample customer records;Recreate o

SYSTEM: Cargo-Tracker

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow registering a new cargo by providing origin, destination, and arrival deadline. - Each registered cargo must receive a unique tracking identifier. - The origin and destination of the cargo must be different. - The origin of the cargo must remain fixed throughout the logistics cycle. - Cargo registration may be completed even before the route is defined. - The arrival deadline must be required and must serve as a criterion for voyage planning. - The system must allow selecting origin and destination only among previously registered logistics locations. - The system must

maintain cargo management as a separate capability, responsible for cargo registration and lifecycle. - The system must retrieve and present candidate routes compatible with the cargo needs. - A candidate route must be considered valid only when it respects the cargo origin, final destination, and arrival deadline. - The user must be able to choose one of the presented routes and assign it to the cargo. - When the cargo does not have a route yet, it must remain identified as "not routed". - When no route meets the informed criteria, the system must inform that there is no feasible option for the current specification. - The system must display the legs of the chosen route, including load points, unload points, voyages, and expected dates. - Route planning depends on origin, destination, and deadline information provided by cargo management. - The system must maintain route planning as an independent capability, responsible for selecting and validating route options. - The system must allow changing the destination of an already registered cargo. - The new destination cannot be equal to the origin or to the current destination. - When the destination changes, the system must reevaluate the current route. - When the current route no longer satisfies the new destination, the cargo must require new routing. - The system must allow replanning routes for cargos that are off the expected route or whose destination has changed. - The system must allow consulting a cargo by tracking identifier. - The consultation must show the current situation of the cargo. - The system must inform the last known location of the cargo. - The system must inform the estimated arrival when the cargo has a valid route and is progressing according to plan. - The system must maintain and display the history of logistics events associated with the cargo. - The history must distinguish expected events from unexpected events. - The system must signal when the cargo is off the expected route. - The system must allow viewing the cargo with geographic support, highlighting origin, final destination, and last known location. - Cargo tracking depends on information maintained by cargo management, route planning, and logistics events. - The system must maintain cargo tracking as a separate capability, responsible for consolidating operational status and history. - The system must allow registering logistics events of the following types: reception, load, unload, customs, and pickup. - Every logistics event must be associated with an existing cargo. - Every logistics event must inform the location where it occurred and the occurrence date. - Load and unload events must be associated with the corresponding voyage. - Events that are not load or unload must not require voyage identification. - The system must reject registration attempts with nonexistent cargo, location, or voyage. - A valid event registration must update the operational situation of the cargo. - After each valid event, the system must recalculate information such as transport status, last known location, next expected step, and plan adherence. - When an event indicates execution different from the planned route, the cargo must be marked as off route. - When the cargo is unloaded at the final destination, the system must mark it as arrived. - When pickup by the recipient is registered, the cargo must be considered complete from a logistics perspective. - Event registration depends on the existence of cargo and voyage maintained by other capabilities. - The system must maintain logistics event registration as a separate capability, responsible for processing and validating received events. - The system

must allow logistics event registration through more than one operational channel. - Supported channels must include at least an operational interface, an integration service, and input file processing. - The system must provide a consolidated view of cargos for monitoring channels. - Integration channels depend on logistics events and cargo tracking. - The system must maintain integration as a separate capability, responsible for receiving events and providing standardized information. - The system must preserve consistency among cargo registration, planned route, and executed events. - No invalid event may change the cargo state. - Origin, destination, deadline, and route compatibility rules must be applied uniformly across all operational channels. - Cargo status must be recalculated whenever there is a relevant planning or execution change. - The system must maintain enough history to reconstruct the cargo logistics cycle. - Each event must keep at least the moment it occurred and the moment it was registered. - Rejected registration attempts must be identifiable for later operational analysis. - The system must allow distinguishing correctly delivered cargos, cargos still in transit, and cargos with operational deviation. - Cargo status updates must occur in a timely manner for operational tracking. - Event reception must continue even when detailed status recalculation occurs in the background. - File inputs must be processed periodically, without depending on constant manual action. - Monitoring channels must reflect the most recent available information about the cargo. - Information presented to the user must be organized by business situation, not only by technical structure. - Operational navigation must facilitate frequent tasks such as locating cargo, consulting details, choosing a route, and registering an event. - The system must reduce operation errors through controlled lists of locations, voyages, event types, and cargo identifiers. - Error messages and missing route alerts must be understandable for administrative and logistics use. - Failures in an isolated registration attempt must not compromise the processing of other cargos. - Invalid input records must be separated for later handling without interrupting the rest of the workload. - The system must treat off-route cargos as visible operational exceptions to allow fast corrective action. - The system must maintain consistency among data presented in the administrative interface, tracking, mobile channels, and integrations. - Main cargo information must be exchangeable in standardized formats to facilitate integration with other systems. - The tracking model must be unique regardless of the channel used to register or consult the cargo.

Reference Services

Cargo Management Service Logistics Network Service Route Planning Service Logistics Event Service Cargo Tracking Service Integration Service

Reference Interactions

Cargo Management Service <-> Logistics Network Service Cargo Management Service <-> Route Planning Service Cargo Management Service <-> Cargo Tracking Service Route Planning Service <-> Logistics Network Service Route Planning Service <-> Cargo Tracking Service Logistics Event Service <-> Cargo Management Service Logistics Event Service <-> Logistics Network Service Logistics Event Service <-> Route Planning Service Logistics Event Service <-> Cargo Tracking Service Integration Service <-> Logistics Event Service

Integration Service <-> Cargo Tracking Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Cargo Management Service,Register new cargo with origin,destination

Route Planning Service,Retrieve and present candidate routes compatible with cargo specifications;Validate candi

Logistics Event Registration Service,Register logistics events of reception,load

Cargo Tracking Service,Consult cargo status and history by tracking identifier;Consolidate transport status,last

Integration Service,Receive logistics events through operational interfaces,integration services

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Cargo Management Service,Register new cargo with origin destination and arrival deadline;Assign unique tracking

Route Planning Service,Select and validate candidate route options;Ensure route compatibility with origin destin

Cargo Tracking Service,Consolidate operational status and event history;Provide cargo consultation by tracking i

Logistics Event Registration Service,Receive and process logistics events including reception load unload custom

Integration Service,Provide event ingestion across operational interface integration service and batch file proc

Location Management Service,Maintain registered logistics locations;Provide controlled list of valid logistics l

Voyage Management Service,Maintain voyages and carrier movement schedules;Provide voyage validation for route pl

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Integration Service,Provide event ingestion across operational interface integration service and batch file proc

Location Management Service,Maintain registered logistics locations;Provide controlled list of valid logistics l

Voyage Management Service,Maintain voyages and carrier movement schedules;Provide voyage validation for route pl

SYSTEM: DayTrader7

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow registering new investors. - Investor registration must include at least user identification, password, password confirmation, full name, address, email, card, and initial opening balance. - The system must validate that the password and password confirmation are identical. - The system must not allow registering an investor with an already existing identification. - The system must authenticate the

investor using username and password. - The system must allow starting a session for authenticated investors. - The system must allow ending the investor session through logout. - The system must prevent access to account, portfolio, and trading resources when the investor is not authenticated. - The system must maintain authentication as a separate capability, responsible exclusively for login, session, and logout. - The system must maintain an individual account for each registered investor. - The investor account must record current balance, initial balance, creation date, last access date, total logins, and total logouts. - The system must allow consulting the investor registration and financial data. - The system must allow updating the investor profile. - The profile update must include at least password, full name, address, email, and registered card. - The system must require filling in the essential fields to complete the profile update. - The system must prevent profile updates when the password and password confirmation are different. - The system must maintain the investor account as a separate capability from authentication, responsible for registration and financial data. - The system must allow querying asset quotes by their codes. - The system must allow simultaneous queries for more than one asset. - For each queried asset, the system must show information such as company name, traded volume, price range, opening price, current price, and variation. - The system must provide a consolidated market summary. - The consolidated market summary must show at least overall performance index, opening index, traded volume, top gaining assets, and top losing assets. - The system must handle invalid asset codes without interrupting the investor navigation. - The system must maintain market data as an independent capability, responsible for quotes and the consolidated summary. - The system must allow buying assets by informing the asset code and desired quantity. - The system must allow selling assets already existing in the investor portfolio. - Every buy or sell operation must generate an individual trading order. - Every order must have its own identifier, type, quantity, price, opening date, completion date, status, and operation fee. - The system must apply a fixed operation fee to buy and sell orders. - For buy operations, the system must debit from the investor balance the amount corresponding to the traded asset plus the operation fee. - For sell operations, the system must credit the investor balance with the net value of the operation, considering the applicable fee discount. - The system must allow orders to be processed immediately or later, according to the simulation operational configuration. - The system must track order progress through business status, including at least open, processing, completed, and cancelled. - The system must not allow the same order to be completed more than once. - The system must cancel a sale when the informed position has already been traded or is no longer available. - The system must show the investor confirmation that the order was sent for processing. - The system must maintain trading as a separate capability, responsible for the lifecycle of buy and sell orders. - The system must maintain the investor portfolio with positions resulting from completed purchases. - When a buy order is completed, the system must register the new position in the investor portfolio. - When a sell order is completed, the system must remove the corresponding position from the portfolio. - The system must allow consulting the investor portfolio. - For each portfolio position, the system must show at least

position identifier, purchase date, asset, quantity, purchase price, current price, invested amount, market value, and gain or loss. - The system must inform when the investor portfolio is empty. - The system must allow selling a position directly from the portfolio. - The system must calculate the total invested amount, current market value, and consolidated gain or loss. - The system must show the investor a consolidated financial position summary, including cash balance, number of positions, portfolio market value, and accumulated result compared to the initial balance. - The system must maintain the portfolio as a separate capability from trading, responsible for positions and consolidated results. - The system must maintain the history of orders associated with the investor. - The system must show a summarized view of the investor's most recent orders. - The system must allow an expanded consultation of order history. - The system must alert the investor about completed orders that have not yet been presented. - The alert must show at least order identifier, status, relevant dates, type, asset, quantity, and applied fee. - The system must update the traded volume and market indicators after buy and sell operations. - The system must reflect these updates in later quote queries and in the consolidated market summary. - The system must allow configuring simulation operational parameters. - The system must allow defining order processing mode, maximum number of simulated users, maximum number of simulated assets, and market summary refresh interval. - The system must allow enabling or disabling completed order alerts. - The system must allow preparing or rebuilding the simulation database. - The system must allow resetting the simulation operational data. - When resetting data, the system must provide summarized execution statistics, including users, assets, orders, cancellations, and existing positions. - The system must maintain simulation administration as a separate capability, responsible for configuration and environment reset. - The system must require prior authentication to access restricted investor resources. - The system must ensure that operations executed during the session are associated with the authenticated investor. - The system must properly end the investor session at logout, reducing the risk of later misuse. - The system must preserve consistency among balance, orders, quotes, and portfolio after each operation. - The system must maintain minimum order lifecycle traceability through identifiers, dates, and status. - The system must handle processing failures without compromising investor data integrity. - The system must prevent, whenever possible, duplicate completion or improper reuse of orders and positions. - The system must support immediate or later order processing to meet different simulation operational strategies. - The system must allow configurable market summary updates to balance information freshness and operational cost. - The system must support operation with high volumes of simulated users, assets, and orders. - The system must present business information clearly, objectively, and organized by usage context. - The system must offer understandable messages for situations such as login failure, registration error, password mismatch, missing required data, and invalid operations. - The system must allow simple navigation among the main views of home, account, portfolio, quotes, orders, and market summary. - The system must provide administrative mechanisms for preparation, configuration, and controlled reset of the simulation environment. - The

system must provide summarized execution indicators to support operational monitoring of the simulation.

Reference Services

Authentication Service Investor Account Service Market Data Service Trading Service Portfolio Service Simulation Administration Service

Reference Interactions

Authentication Service <-> Investor Account Service Trading Service <-> Authentication Service Trading Service <-> Investor Account Service Trading Service <-> Market Data Service Trading Service <-> Portfolio Service Portfolio Service <-> Investor Account Service Portfolio Service <-> Market Data Service Simulation Administration Service <-> Authentication Service Simulation Administration Service <-> Investor Account Service Simulation Administration Service <-> Market Data Service Simulation Administration Service <-> Trading Service Simulation Administration Service <-> Portfolio Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Authentication Service,Authenticate investor credentials;Manage investor session lifecycle;Terminate sessions on
Investor Account Service,Register new investors;Validate investor registration and profile data;Maintain investo
Market Data Service,Provide asset quotes queries;Generate consolidated market summary;Update traded volume and m
Trading Service,Generate buy and sell trading orders;Apply fixed operation fees;Track order progress and status
Portfolio Service,Maintain investor asset positions;Register positions from completed buy orders;Remove position
Simulation Administration Service,Configure simulation operational parameters;Configure order processing modes a

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Authentication Service,Authenticating investors using username and password;Managing investor login sessions and
Investor Account Service,Registering new investors and validating unique identification;Maintaining investor acc
Market Data Service,Querying individual and simultaneous asset quotes;Generating consolidated market summaries a
Trading Service,Managing buy and sell order lifecycles and statuses;Applying operation fees to orders;Debiting a
Portfolio Service,Maintaining investor portfolio positions;Calculating current market values and gains or losses
Simulation Administration Service,Configuring simulation operational parameters;Preparing and rebuilding simulat

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Trading Service,Generate buy and sell trading orders;Apply fixed operation fees;Track order progress and status

Portfolio Service,Maintain investor asset positions;Register positions from completed buy orders;Remove position
Simulation Administration Service,Configure simulation operational parameters;Configure order processing modes a

SYSTEM: Jokul

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow user registration with username and password. - The system must allow login for already registered users. - The system must differentiate at least two usage profiles: regular user and administrator. - The system must deny access when the user does not exist, when the password is incorrect, or when the informed profile does not match the registration. - The system must maintain the logged user identification during browser navigation. - The system must show the resource management area only to users identified as administrators. - The system must allow the user to end the session. - The system must operate with separation between regular user and administrator profiles. - Collection management functions must be restricted to the administrator profile. - The user session must remain identified in the browser until logout or local data clearing. - The system must prevent duplicate user registration with the same username. - The system must maintain authentication as a separate capability, responsible exclusively for registration, login, session, and profile control. - The system must display a home page with the list of available movies. - Each listed movie must show at least title, poster, and rating. - The system must organize the movie listing into pages. - The system must allow consulting the total number of registered movies to support paginated navigation. - The system must allow opening the movie details page from the listing. - The system must present a details page for each movie. - The details page must show at least title, rating, alternative name, release date, duration, direction, screenplay, cast, synopsis, poster, and categories. - The system must allow viewing the full movie cast. - The system must prevent consulting details of nonexistent movies, returning an appropriate error message. - The system must provide a navigation area for movie categories. - The system must allow viewing all registered categories. - The system must allow listing only movies belonging to a selected category. - A movie must be able to belong to more than one category. - The system must allow returning from the category view to the general catalog view. - The system must maintain the movie catalog as a separate capability, responsible for listing, details, and categories. - The catalog capability depends on authentication to validate user access and restrict management operations. - The system must allow starting movie playback from the details page. - The system must provide a download link for the movie file when the resource is ready for consumption. - Before playback, the system must ensure that the video file is available in the area used by the application for content delivery. - If the file is not yet available in the delivery area, the system must try to obtain it from the configured local repository. - The standard playback process assumes that the physical movie file corresponds to the registered title. - Video file upload and movie registration are related but independent steps. - Automatic playback depends on the existence of an MP4 file corresponding to the registered title. - The

source movie filename must follow the expected pattern configured in the environment for automatic loading to work. - The system must support large video file uploads, up to 2 GB per file. - The operational process must consider MP4 as the standard video format. - The application environment must have a local area to temporarily store or serve the video files used during playback. - The application environment must have a configured local repository from which movies can be loaded on demand. - The system must maintain media as a separate capability from the catalog, responsible for upload, storage, and video delivery. - The media capability depends on the catalog to identify the registered movie corresponding to each video file. - The media capability depends on authentication to validate access before playback and download. - The administrator must be able to register a new movie in the catalog. - Movie registration must include at least title, rating, alternative name, release date, duration, direction, screenplay, cast, categories, synopsis, and poster. - The system must prevent duplicate movie registration with the same title. - The administrator must be able to upload the movie video file to the application. - The administrator must be able to delete a movie from the catalog. - The system must allow batch deletion of more than one movie. - The system must inform the administrator which movies were selected before batch deletion. - The system must prevent deleting nonexistent movies. - The movie title must be treated as the unique identifier in the catalog registration. - The application must be accessible through a web interface. - Catalog navigation must be simple and direct, with visible access to the home page, categories, details, and playback. - The application must present clear success, warning, and failure messages for main operations such as login, registration, inclusion, deletion, upload, and playback. - Catalog display must be paginated in blocks of up to 12 movies per page, supporting visual organization and perceived performance. - The system must maintain a single response pattern for operations, indicating status, message, and returned data. - User, movie, and category information must be durably persisted. - The system must preserve user and movie uniqueness to avoid duplicates in the collection and access. - Error handling must return controlled messages to the user whenever possible. - User, movie, and category data must be stored in a structured database that supports catalog relationships. - The solution must allow the user interface and application services to evolve separately. - The user interface must be able to consume application services even when published in a different environment. - The intention to restrict resource management to the administrator is clear in the interface, but this restriction is not completely validated in the backend. - The current system is more oriented to a local movie collection than to remote storage or an external streaming service. - Playback availability depends on both movie registration and the existence of the corresponding physical file.

Reference Services

Authentication Service Movie Catalog Service Media Service

Reference Interactions

Movie Catalog Service <-> Authentication Service Media Service <-> Authentication Service Media Service <-> Movie Catalog Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Authentication Service,Register users with unique usernames;Authenticate users and maintain sessions;Manage regu

Catalog Service,List available movies with pagination;Display movie details and cast information;Manage categori

Media Service,Handle video file uploads;Manage video storage and staging in the content delivery area;Obtain vid

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Authentication Service,User registration with username and password;User authentication and session management;U

Catalog Service,Movie catalog listing and pagination;Movie details and cast presentation;Category management and

Media Service,Video file upload processing;Video storage and local repository management;Content delivery area m

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

This consolidated architecture is the implementation recommendation produced after considering the intermediate

Consolidated Architecture CSV

Microservice,Responsibilities,Communicates With

Authentication Service,Register users with unique usernames;Authenticate users and maintain sessions;Manage regu

Catalog Service,List available movies with pagination;Display movie details and cast information;Manage categori

Media Service,Handle video file uploads;Manage video storage and staging in the content delivery area;Obtain vid

SYSTEM: JpetStore

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must allow visitors and customers to enter the virtual store and browse the main catalog. - The system must organize products by animal categories, including fish, dogs, reptiles, cats, and birds. - The system must allow access to categories through different navigation paths, such as the main menu, quick shortcuts, and visual elements on the home page. - The system must allow viewing the product list of each category. - The system must allow viewing the available items of each product. - The system must allow viewing item details, including description, identifier, price, and stock status.
- The system must allow searching products by keyword. - The search must consider the product name as the basis for returning results. - When no keyword is provided, the system must block the search and guide the user to fill in the search field. - The system must maintain the catalog as a separate capability, responsible for products, categories, items, and search. - The catalog capability depends on inventory to provide updated stock information for items. - The system must allow registering new customers by filling in identification, contact, address, and access credential data. - The system must allow customers to authenticate using username and password. - The system must inform when authentication cannot be completed due to invalid username or password. - The system must allow customers to end their usage session. - The system must allow

customers to edit their registration data and profile preferences. - The system must allow customers to define their preferred language. - The system must allow customers to choose a favorite category. - The system must allow customers to enable a personalized product list based on their favorite category. - The system must allow customers to enable a promotional banner related to their favorite category. - The system must reflect customer preferences, such as the personalized list and banner associated with the favorite category. - The system must maintain the customer account as a separate capability, responsible for registration, profile, and preferences. - The system must maintain authentication as a capability separate from the customer account, responsible for login, session, and access control. - The customer account capability depends on authentication to validate user access. - The system must allow adding items to the shopping cart from product pages and item detail pages. - When the same item is added more than once, the system must consolidate the item in the cart and increase its quantity. - The system must allow viewing the cart with selected items, quantities, unit prices, and totals by item. - The system must inform, for each cart item, whether it is available in stock at the time of inclusion. - The system must allow updating item quantities in the cart. - The system must allow removing items from the cart. - When the informed quantity for an item is less than 1, the system must remove it from the cart. - The system must automatically calculate the cart subtotal based on selected items and quantities. - The system must maintain the cart as a separate capability, responsible for the temporary selection of items. - The shopping cart capability depends on the catalog to identify the selected items. - The shopping cart capability depends on inventory to validate item availability. - The system must inform stock availability when viewing an item. - When there is no available quantity in stock, the system must indicate that the item is on backorder. - The system must consult stock status in an updated manner at critical points of the purchase journey. - When completing an order, the system must register the decrease of the purchased quantity from stock. - The system must maintain inventory as a separate capability, responsible for item availability and stock decrease. - The inventory capability depends on the catalog to identify the items whose stock is being managed. - The system must require the user to be authenticated to start the checkout process. - When starting an order, the system must use the customer registration data to initially fill in billing and delivery information. - The system must use the cart items to automatically compose the order. - The system must allow customers to inform or adjust order payment data. - The system must allow choosing the card type among the options available in the application. - The system must allow customers to indicate a delivery address different from the billing address. - When the customer chooses a different delivery address, the system must present a specific step for filling in that address. - Before finalization, the system must present a confirmation step with the informed billing and delivery data. - The system must complete the order only after explicit customer confirmation. - When completing the order, the system must generate a unique purchase identifier. - When completing the order, the system must record the purchase date, acquired items, quantities, values, and initial order status. - After purchase completion, the system must clear the customer

cart. - After purchase completion, the system must display a confirmation message for order receipt. - The system must maintain the order as a separate capability, responsible for checkout and order lifecycle. - The order capability depends on authentication to validate access. - The order capability depends on the customer account to obtain registration data. - The order capability depends on the shopping cart to obtain the items being purchased. - The order capability depends on inventory to register the stock decrease. - The system must allow customers to consult their list of completed orders. - The order listing must show at least order number, date, and total value. - The system must allow consulting the details of a specific order. - The order details must show payment data, addresses, carrier, status, purchased items, quantities, prices, and total purchase value. - The system must restrict order viewing to the respective customer responsible for the purchase. - The system must protect checkout and order consultation features through customer authentication. - The system must properly end the customer session at logout, avoiding continued unauthorized authenticated access. - The system must not unnecessarily expose the customer password during navigation after login. - The system must maintain consistency among cart, order, and inventory, avoiding divergences between the completed purchase and registered items. - The system must ensure that the order is registered with all main elements, including header, status, and purchased items. - The system must ensure uniqueness in order numbering, allowing purchase traceability. - The system must automatically calculate cart and order values, reducing manual error risk. - The system must handle invalid inputs in a controlled way, displaying appropriate messages without compromising application continuity. - The system must prevent inconsistent operations, such as starting checkout without authentication or silently manipulating nonexistent items. - The system must preserve order history for later customer consultation. - The system must offer simple and understandable navigation, with clear access to categories, search, cart, customer account, and help. - The purchase process must be conducted in logical and progressive steps, making checkout easier to understand. - The system must provide guidance and confirmation messages at important flow points, such as login failure, empty search field, and completed order.

Reference Services

Authentication Service Customer Account Service Catalog Service Inventory Service Shopping Cart Service Order Service

Reference Interactions

Customer Account Service <-> Authentication Service Customer Account Service <-> Catalog Service Catalog Service <-> Inventory Service Shopping Cart Service <-> Catalog Service Shopping Cart Service <-> Inventory Service Order Service <-> Authentication Service Order Service <-> Customer Account Service Order Service <-> Shopping Cart Service Order Service <-> Inventory Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Catalog Service,Manage animal categories and product listings;Display item details and descriptions;Search produ

Inventory Service,Track item stock availability;Manage backorder status indications;Register stock decreases upo

Customer Account Service,Register new customer accounts and access credentials;Manage customer profiles and cont

Authentication Service,Authenticate user credentials;Manage customer login sessions and logout;Validate access t

Shopping Cart Service,Manage temporary item selections;Consolidate item quantities and calculate cart subtotals;

Order Service,Conduct checkout process and compose orders;Collect payment information and delivery addresses;Gen

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Catalog Service,Manage products categories and items;Browse catalog and view item details;Search products by key

Customer Account Service,Manage customer registration and profile data;Manage customer preferences and favorite

Authentication Service,Authenticate customers with username and password;Manage user sessions;Handle logout and

Shopping Cart Service,Manage temporary item selection;Consolidate and update item quantities;Calculate cart subt

Inventory Service,Manage item availability;Provide updated stock status and indicate backorders;Register stock d

Order Service,Coordinate checkout process;Assemble order from cart and customer registration data;Manage payment

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

This consolidated architecture is the implementation recommendation produced after considering the intermediate

Consolidated Architecture CSV

Microservice,Responsibilities,Communicates With

Catalog Service,Manage animal categories and product listings;Display item details and descriptions;Search produ

Inventory Service,Track item stock availability;Manage backorder status indications;Register stock decreases upo

Customer Account Service,Register new customer accounts and access credentials;Manage customer profiles and cont

Authentication Service,Authenticate user credentials;Manage customer login sessions and logout;Validate access t

Shopping Cart Service,Manage temporary item selections;Consolidate item quantities and calculate cart subtotals;

Order Service,Conduct checkout process and compose orders;Collect payment information and delivery addresses;Gen

SYSTEM: Pet-Clinic

SECTION 1 - INPUTS PROVIDED

System Requirements

- The system must provide a home page for user entry. - The system must offer navigation to the main functionalities for owners and veterinarians. - The system must have a simple web interface, with easy-to-understand forms and navigation. - The system must present clear success, error, and validation messages. - The system must be usable on different screen sizes, such as desktop and mobile. - The system must support multiple interface languages. - Messages and labels must remain consistent across supported languages. - The system must display understandable error pages for known and unexpected failures. - The system must allow registering owners with first name, last name, address, city, and telephone. - The system must allow updating the data of an already registered owner. - The system must allow searching owners by the beginning of the informed last name. - The system must allow broad owner search when no last name is

provided. - The system must direct directly to the detailed owner registration when the search returns only one result. - The system must present a paginated list when the search returns several owners. - The system must inform the user when no owner match is found. - The system must display the complete data of the selected owner. - The system must maintain owner management as a separate capability, responsible exclusively for owner data and consultation. - The system must allow registering an animal linked to an owner. - The system must allow updating the data of an already registered animal. - The system must record, for each animal, name, type, and birth date. - The system must provide a predefined list of animal types for selection. - The system must prevent the same owner from having two animals with the same name. - The system must prevent registering a future birth date for the animal. - The system must maintain animal management as a separate capability, responsible for animal data and the link with the owner. - The animal management capability depends on owner management to validate the owner link. - The system must allow registering a visit for a specific animal. - The visit record must contain at least date and service description. - The system must suggest the current date as the initial value for a new visit. - The system must associate the registered visit with the animal history and the respective owner. - The system must display the visit history of the animals linked to the owner. - Visit registration must remain consistent even when veterinarian assignment occurs in a separate step. - The system must resume pending internal processes after restart, avoiding loss of relevant events. - The system must maintain visit management as a separate capability, responsible for visit registration and service history. - The visit management capability depends on animal management to validate the animal. - The visit management capability depends on veterinarian management to assign the visit. - The system must list registered veterinarians with their names and specialties. - The system must inform the next expected visit for each veterinarian. - The system must present the veterinarian list in a paginated manner. - The system must automatically assign each new visit to a veterinarian. - Automatic assignment must prioritize the veterinarian with the lowest future workload. - The system must periodically disregard past visit assignments when calculating veterinarian workload. - The system must provide the veterinarian list in a structured format for consumption by other systems. - Frequent veterinarian queries must be optimized to reduce response time. - The system must maintain veterinarian management as a separate capability, responsible for veterinarian data, specialties, and workload. - The veterinarian management capability depends on visit management to calculate workload and next visits. - The system must validate required data before completing registrations and updates. - The system must not allow direct modification of internal identifiers through forms. - Static interface resources must be temporarily reusable by the browser to improve loading. - The system must support growth in the volume of owners and veterinarians without compromising basic navigation, using pagination. - The system must be able to operate with different data storage configurations. - The system must be organized to facilitate isolated evolution of owners, veterinarians, and general functions. - The main system behaviors must be verifiable through automated tests.

Reference Services

Owner Management Service Pet Management Service Visit Management Service Veterinarian Management Service

Reference Interactions

Pet Management Service <-> Owner Management Service Visit Management Service <-> Pet Management Service Visit Management Service <-> Veterinarian Management Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Owner Service,Register owners;Update owner details;Search owners by last name;Retrieve owner information,

Animal Service,Register animals linked to owners;Update animal details;Provide predefined animal types;Validate

Visit Service,Register animal visits;Maintain visit and service history;Associate visits with animal history;Man

Veterinarian Service,List registered veterinarians and specialties;Calculate veterinarian workload and next expe

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Owner Management Service,Register owners with contact details;Update registered owner records;Search owners by l

Animal Management Service,Register animals linked to owners;Update registered animal records;Record animal name

Visit Management Service,Register visits for specific animals with date and service description;Suggest current

Veterinarian Management Service,Manage registered veterinarians with names and specialties;Display paginated vet

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

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Consolidated Architecture CSV

Microservice,Responsibilities,Communicates With

Owner Management Service,Register owners with contact details;Update registered owner records;Search owners by l

Animal Management Service,Register animals linked to owners;Update registered animal records;Record animal name

Visit Management Service,Register visits for specific animals with date and service description;Suggest current

Veterinarian Management Service,Manage registered veterinarians with names and specialties;Display paginated vet

SYSTEM: TNTConcept

SECTION 1 - INPUTS PROVIDED

System Requirements

1. Administration and Configuration - The system must allow maintaining support records used by other areas. - These records include departments, contract types, user categories, organization types, interaction types, tax reasons, billing regimes, and frequencies. - The system must allow individual user configurations. - This capability provides basic parameters to identity, workforce, CRM, billing, and notification areas.

2. Identity and Access - The system must allow registering and maintaining users with personal, contractual, department, category, access profile, and work agreement data. -

The system must allow password change, administrative reset, and forced renewal. - The system must control access by user profile and scope. - This capability depends on administration to obtain user categories and general parameters. - It provides user validation to workforce, project tracking, content management, and other areas. ## 3. Workforce Management - The system must allow maintaining work agreements and vacation requests. - It must control vacation request states and calculate balances. - This capability depends on identity to validate employees. - It depends on administration to obtain contract types and work rules. - It sends notifications for vacation approvals and contract expiration. ## 4. Project and Work Tracking - The system must allow maintaining projects linked to clients. - It must register internal and external activities. - It must classify activities and calculate worked hours. - It must support evidence policies. - This capability depends on workforce to validate employees. - It depends on identity to validate access. - It depends on file storage to attach evidence. ## 5. CRM and Proposals - The system must allow maintaining organizations, contacts, interactions, and proposals. - It must control proposal states and composition. - This capability depends on administration for organization and interaction types. - It depends on project tracking to link proposals to projects. ## 6. Financial Management - The system must maintain accounts, entries, and financial indicators. - It depends on billing to obtain invoice data. - It depends on project tracking to import billable hours and costs. ## 7. Billing - The system must detail invoices and calculate totals, taxes, and balances. - It depends on CRM for proposal conversion. - It depends on project tracking for billable hours. - It depends on financial management for entries and payments. - It depends on file storage for attachments. ## 8. Content and Document Management - The system must organize documents by categories and versions. - It depends on file storage. - It depends on identity to associate documents with users and areas. ## 9. Commissioning - The system must register and manage orders/commissions. - It depends on project tracking, workforce, file storage, and notification. ## 10. Reporting - The system must consolidate and export reports. - It depends on all main business domains. ## 11. Notification - The system must send email notifications. - It depends on administration for settings. - It is used by workforce, project tracking, commissioning, and other areas. ## 12. File Storage - The system must store and retrieve files. - It depends on administration for storage settings.

Reference Services

Administration and Configuration Service Identity and Access Service Workforce Service CRM Service Proposal Service Project Service Work Tracking Service Financial Management Service Billing Service Content Management Service Commissioning Service Reporting Service Notification Service File Storage Service

Reference Interactions

Identity and Access Service <-> Administration and Configuration Service Identity and Access Service <-> Notification Service Workforce Service <-> Administration and Configuration Service Workforce Service <-> Identity and Access Service Workforce Service <-> Content Management Service Workforce Service <-> Notification Service CRM Service <-> Administration and Configuration Service CRM Service <-> Proposal Service

CRM Service <-> Project Service Proposal Service <-> Billing Service Work Tracking
Service <-> Project Service Work Tracking Service <-> Workforce Service Work Tracking
Service <-> File Storage Service Work Tracking Service <-> Notification Service Billing
Service <-> CRM Service Billing Service <-> Project Service Billing Service <-> Work
Tracking Service Billing Service <-> Financial Management Service Billing Service <->
File Storage Service Content Management Service <-> File Storage Service Commissioning
Service <-> Project Service Commissioning Service <-> Workforce Service Commissioning
Service <-> File Storage Service Commissioning Service <-> Notification Service
Reporting Service <-> Workforce Service Reporting Service <-> CRM Service Reporting
Service <-> Proposal Service Reporting Service <-> Project Service Reporting Service <->
Work Tracking Service Reporting Service <-> Billing Service Reporting Service <->
Financial Management Service Reporting Service <-> Commissioning Service Reporting
Service <-> File Storage Service Notification Service <-> Administration and
Configuration Service File Storage Service <-> Administration and Configuration Service

SECTION 2 - AGENT PROPOSALS

Agent Proposals

Agent A (Software Architect)

Microservice,Responsibilities,Communicates With

Administration and Configuration Service,Maintain system support records;Manage user individual configurations;P
Identity and Access Service,Register and maintain user profiles;Manage passwords and credential renewals;Enforce
Workforce Management Service,Maintain work agreements and vacation requests;Control vacation request states and
Project and Work Tracking Service,Maintain client-linked projects;Register and classify internal and external ac
CRM and Proposals Service,Maintain organizations contacts and interactions;Manage commercial proposals and propo
Financial Management Service,Maintain financial accounts and entries;Calculate and monitor financial indicators,
Billing Service,Generate and detail invoices;Calculate invoice totals taxes and balances;Process proposal conver
Content and Document Management Service,Organize documents by categories and versions;Associate documents with u
Commissioning Service,Register and manage orders and commissions,Workforce Management Service;Project and Work T
Reporting Service,Consolidate cross-domain business data;Generate and export analytical reports,Administration a
Notification Service,Send email notifications;Process notification triggers from business domains,Administration
File Storage Service,Store uploaded files and attachments;Retrieve stored files,Administration and Configuration

Agent B (Communication Specialist)

Microservice,Responsibilities,Communicates With

Administration Service,Maintain support records including departments and contract types and user categories;Man
Identity and Access Service,Register and maintain user personal and contractual and access data;Manage password
Workforce Management Service,Maintain work agreements and vacation requests;Control vacation request states and
Project Tracking Service,Maintain projects linked to clients;Register internal and external activities;Classify
CRM and Proposals Service,Maintain organizations and contacts and interactions;Manage commercial proposals and p
Financial Management Service,Maintain financial accounts and entries and indicators;Import billable hours and co
Billing Service,Detail invoices and calculate totals and taxes and balances;Convert proposals into billings;Proc
Content Management Service,Organize documents by categories and versions;Associate documents with users and busi
Commissioning Service,Register and manage orders and commissions;Track commission workflows and lifecycle,File S

Reporting Service,Consolidate data across business domains;Generate and export consolidated business reports,Adm

Notification Service,Send email notifications;Process notification triggers from business workflows,Administrati

File Storage Service,Store and retrieve files and evidence attachments;Manage storage operations,Administration

SECTION 3 - FINAL ARCHITECTURE RECOMMENDED BY THE PIPELINE

Final Architecture Recommended by the Pipeline

This consolidated architecture is the implementation recommendation produced after considering the intermediate

Consolidated Architecture CSV

Microservice,Responsibilities,Communicates With

Administration and Configuration Service,Maintain system support records;Manage user individual configurations;P

Identity and Access Service,Register and maintain user profiles;Manage passwords and credential renewals;Enforce

Workforce Management Service,Maintain work agreements and vacation requests;Control vacation request states and

Project and Work Tracking Service,Maintain client-linked projects;Register and classify internal and external ac

CRM and Proposals Service,Maintain organizations contacts and interactions;Manage commercial proposals and propo

Financial Management Service,Maintain financial accounts and entries;Calculate and monitor financial indicators,

Billing Service,Generate and detail invoices;Calculate invoice totals taxes and balances;Process proposal conver

Content and Document Management Service,Organize documents by categories and versions;Associate documents with u

Commissioning Service,Register and manage orders and commissions,File Storage Service;Notification Service;Proje

Reporting Service,Consolidate cross-domain business data;Generate and export analytical reports,Administration a

Notification Service,Send email notifications;Process notification triggers from business domains,Administration

File Storage Service,Store uploaded files and attachments;Retrieve stored files,Administration and Configuration

SECTION 4 - EVALUATION METRICS

Evaluation Metrics

Precision, Recall, and F1-score are calculated by comparing the generated architectures with the reference services and interactions provided in the inputs. The quality of these metrics depends on how accurately the reference data represents the desired architecture.

Services Metrics: System | Proposal | Precision | Recall | F1-Score

7ep | Proposal A | 1.0000 | 1.0000 | 1.0000

7ep | Proposal B | 1.0000 | 1.0000 | 1.0000

7ep | Consolidated | 1.0000 | 1.0000 | 1.0000

acmeair | Proposal A | 1.0000 | 1.0000 | 1.0000

acmeair | Proposal B | 1.0000 | 1.0000 | 1.0000

acmeair | Consolidated | 1.0000 | 1.0000 | 1.0000

Cargo-Tracker | Proposal A | 1.0000 | 0.8333 | 0.9091

Cargo-Tracker | Proposal B | 0.7143 | 0.8333 | 0.7692

Cargo-Tracker | Consolidated | 0.7143 | 0.8333 | 0.7692

DayTrader7 | Proposal A | 1.0000 | 1.0000 | 1.0000

DayTrader7 | Proposal B | 1.0000 | 1.0000 | 1.0000

DayTrader7 | Consolidated | 1.0000 | 1.0000 | 1.0000

Jokul | Proposal A | 1.0000 | 1.0000 | 1.0000

Jokul | Proposal B | 1.0000 | 1.0000 | 1.0000

Jokul | Consolidated | 1.0000 | 1.0000 | 1.0000

JpetStore | Proposal A | 1.0000 | 1.0000 | 1.0000

JpetStore | Proposal B | 1.0000 | 1.0000 | 1.0000

JpetStore | Consolidated | 1.0000 | 1.0000 | 1.0000

Pet-Clinic | Proposal A | 1.0000 | 1.0000 | 1.0000

Pet-Clinic | Proposal B | 1.0000 | 1.0000 | 1.0000

Pet-Clinic | Consolidated | 1.0000 | 1.0000 | 1.0000

TNTConcept | Proposal A | 1.0000 | 0.8571 | 0.9231

TNTConcept | Proposal B | 1.0000 | 0.8571 | 0.9231

TNTConcept | Consolidated | 1.0000 | 0.8571 | 0.9231

Interactions Metrics: System | Proposal | Precision | Recall | F1-Score

7ep | Proposal A | 0.9091 | 1.0000 | 0.9524

7ep | Proposal B | 0.9091 | 1.0000 | 0.9524

7ep | Consolidated | 0.9091 | 1.0000 | 0.9524

acmeair | Proposal A | 1.0000 | 1.0000 | 1.0000

acmeair | Proposal B | 1.0000 | 1.0000 | 1.0000

acmeair | Consolidated | 1.0000 | 1.0000 | 1.0000

Cargo-Tracker | Proposal A | 0.0000 | 0.0000 | 0.0000

Cargo-Tracker | Proposal B | 0.5833 | 0.6364 | 0.6087

Cargo-Tracker | Consolidated | 0.5833 | 0.6364 | 0.6087

DayTrader7 | Proposal A | 1.0000 | 0.8333 | 0.9091

DayTrader7 | Proposal B | 1.0000 | 0.8333 | 0.9091

DayTrader7 | Consolidated | 1.0000 | 0.8333 | 0.9091

Jokul | Proposal A | 0.6667 | 0.6667 | 0.6667

Jokul | Proposal B | 0.6667 | 0.6667 | 0.6667

Jokul | Consolidated | 0.6667 | 0.6667 | 0.6667

JpetStore | Proposal A | 1.0000 | 0.8889 | 0.9412

JpetStore | Proposal B | 1.0000 | 0.8889 | 0.9412

JpetStore | Consolidated | 1.0000 | 0.8889 | 0.9412

Pet-Clinic | Proposal A | 1.0000 | 1.0000 | 1.0000

Pet-Clinic | Proposal B | 1.0000 | 1.0000 | 1.0000

Pet-Clinic | Consolidated | 1.0000 | 1.0000 | 1.0000

TNTConcept | Proposal A | 0.7879 | 0.7429 | 0.7647

TNTConcept | Proposal B | 0.7879 | 0.7429 | 0.7647

TNTConcept | Consolidated | 0.7879 | 0.7429 | 0.7647

SECTION 5 - BEST RESULTS

System | Best Services | Best Interactions

7ep | Proposal A (F1 1.0000) | Proposal A (F1 0.9524)

acmeair | Proposal A (F1 1.0000) | Proposal A (F1 1.0000)

Cargo-Tracker | Proposal A (F1 0.9091) | Proposal B (F1 0.6087)

DayTrader7 | Proposal A (F1 1.0000) | Proposal A (F1 0.9091)

Jokul | Proposal A (F1 1.0000) | Proposal A (F1 0.6667)

JpetStore | Proposal A (F1 1.0000) | Proposal A (F1 0.9412)

Pet-Clinic | Proposal A (F1 1.0000) | Proposal A (F1 1.0000)

TNTConcept | Proposal A (F1 0.9231) | Proposal A (F1 0.7647)

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